Pandas Data Analysis - Reading Files, Basic Analysis, GroupBy, and Binning

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1. Introduction

Pandas is a foundational Python library used for data analysis and manipulation. It provides flexible data structures like `DataFrames` that allow analysts and data scientists to clean, analyze, and visualize data efficiently.

Why it matters: Whether you're exploring survey data, analyzing sales records, or preparing data for machine learning, Pandas helps you read, process, and make sense of large datasets with just a few lines of code.

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2. Learning Objectives

By the end of this lesson, students will be able to:

1. Load data from CSV and Excel files using Pandas.
2. Perform basic data exploration and descriptive statistics.
3. Use `groupby()` to segment and summarize data.
4. Use binning to categorize continuous variables into discrete intervals.

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3. Concept Explanation

Key Concepts

- `DataFrame` and `Series` objects

- Reading files: `read_csv()`, `read_excel()`
- Descriptive statistics: `.head()`, `.describe()`, `.info()`
- `groupby()` for aggregations
- `cut()` and `qcut()` for binning continuous variables

Real-World Applications

- Retail: Grouping sales by product categories.
- Education: Binning students' grades into letter grades.
- Healthcare: Grouping patients by age ranges.

Student Analogy

Think of a Pandas DataFrame like an Excel spreadsheet. Each row is a record (like a student), and each column is a feature (like name, age, marks). Just like you filter, sort, and summarize Excel data, Pandas helps you do the same in Python-only faster and at scale.

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4. Python Code Examples

4.1 Reading Data from a File

python

```python
import pandas as pd

# Load CSV
df = pd.read_csv('students.csv')

# Load Excel
# df = pd.read_excel('students.xlsx')
```

```
# Show the first 5 rows
print(df.head())
```

4.2 Basic Data Analysis

python

```
# Overview of the dataset
print(df.info())

# Summary statistics
print(df.describe())

# Check for missing values
print(df.isnull().sum())
```

4.3 GroupBy Example

python

```
# Group students by 'Grade' and calculate average marks
avg_marks = df.groupby('Grade')['Marks'].mean()
print(avg_marks)
```

4.4 Binning Example

python

```
# Suppose we have a column 'Age'
bins = [0, 12, 18, 25, 60]
labels = ['Child', 'Teen', 'Young Adult', 'Adult']

df['Age_Group'] = pd.cut(df['Age'], bins=bins, labels=labels)
print(df[['Age', 'Age_Group']].head())
```

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5. Common Mistakes & Misunderstandings

Mistake	Clarification
Using `groupby` without `agg()` or `mean()`	GroupBy must be followed by an aggregation
Confusing `cut()` and `qcut()`	`cut()` = equal-sized bins; `qcut()` = quantile-based bins
Forgetting to assign binned column	Always assign the result of `cut()` back to a column
Not checking for missing values	Always check `.isnull().sum()` before analysis

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6. Interactive Exercise

Exercise 1: Analyze Student Performance

Dataset: `students.csv` with columns `Name`, `Age`, `Gender`, `Grade`, `Marks`.

Tasks:

- Load the data.
- Find the average marks per grade.
- Create an Age Group column using bins: `[0, 12, 18, 25]` with labels `['Child', 'Teen', 'Young Adult']`.
- Count how many students fall into each age group.

Exercise 2: Supermarket Sales (Advanced)

Dataset: `supermarket.csv` with columns `CustomerID`, `ProductCategory`, `PurchaseAmount`, `Gender`, `Age`.

Tasks:

- Group by `ProductCategory` and calculate total purchases.

- Bin customers into age groups (`<20`, `20–40`, `40–60`, `60+`) and count frequency.
- Identify the category with the highest average purchase.

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7. Student-Friendly Analogy or Story

Imagine you run a cafeteria. You get a big Excel sheet with daily purchases. Now you want to:

- Read the sheet (Pandas `read_csv`)
- See how many people bought fries (filtering)
- See which age group buys the most sodas (binning)
- Find out which day of the week you sell the most (groupby)

Pandas helps you do all this in minutes, not hours!

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8. Recommended Resources

- Pandas Documentation
- Kaggle: Pandas Micro-Course
- Book: *Python for Data Analysis* by Wes McKinney
- YouTube: Corey Schafer's Pandas tutorials

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9. Wrap-Up/Recap

Today, you learned how to:

- Import data using Pandas

- Explore data using `.head()`, `.describe()`
- Summarize with `groupby()`
- Categorize continuous data using `cut()` for better insight

These tools form the foundation of most data analysis workflows in the real world.

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10. Quiz or Reflection Questions

1. What function is used to read CSV files in Pandas?
2. How does `groupby()` work in Pandas?
3. What's the difference between `cut()` and `qcut()`?
4. Why is binning useful in data analysis?